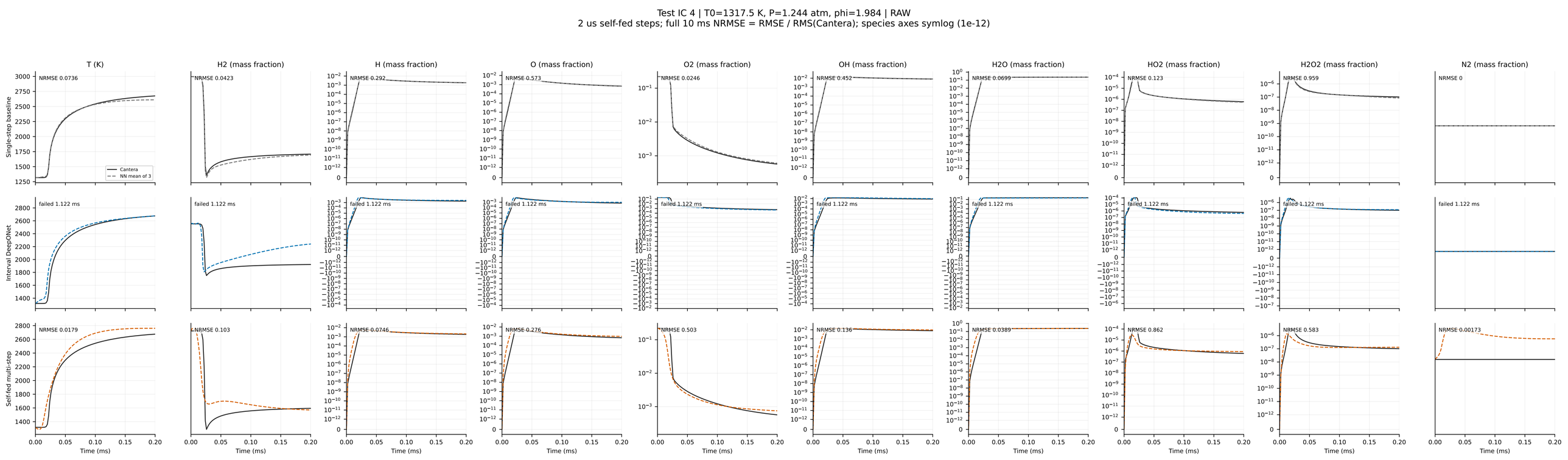
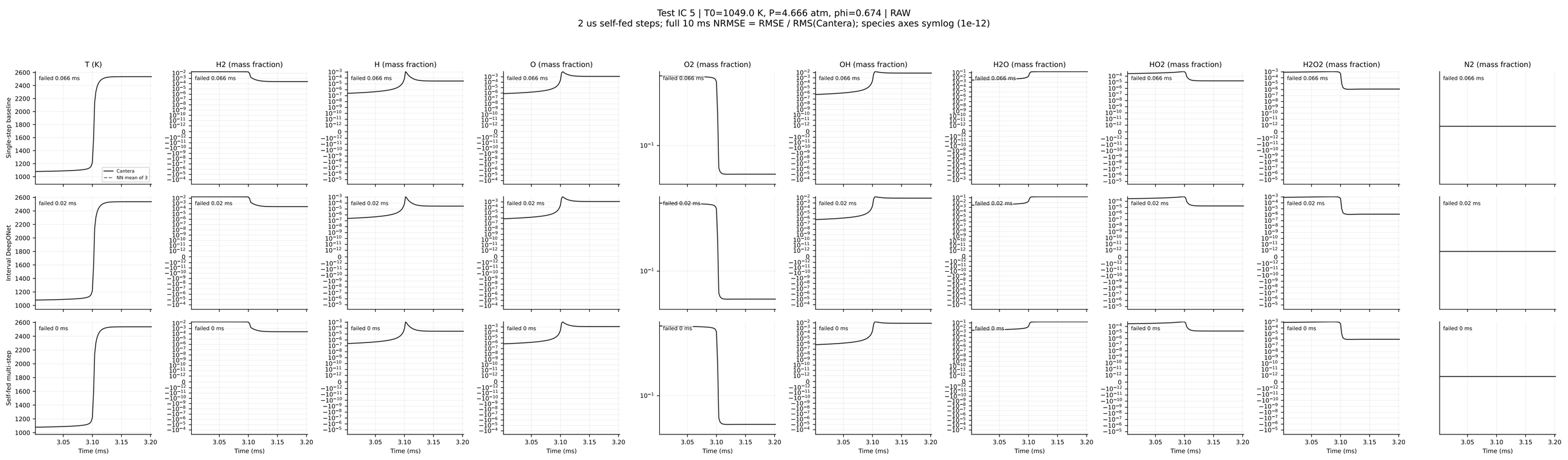
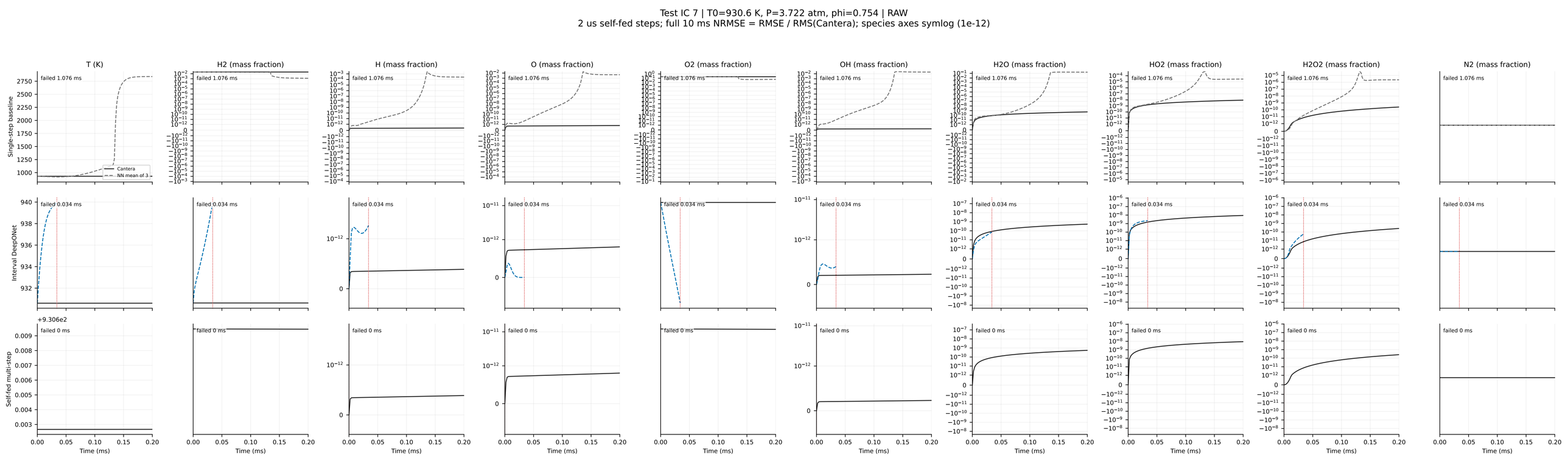
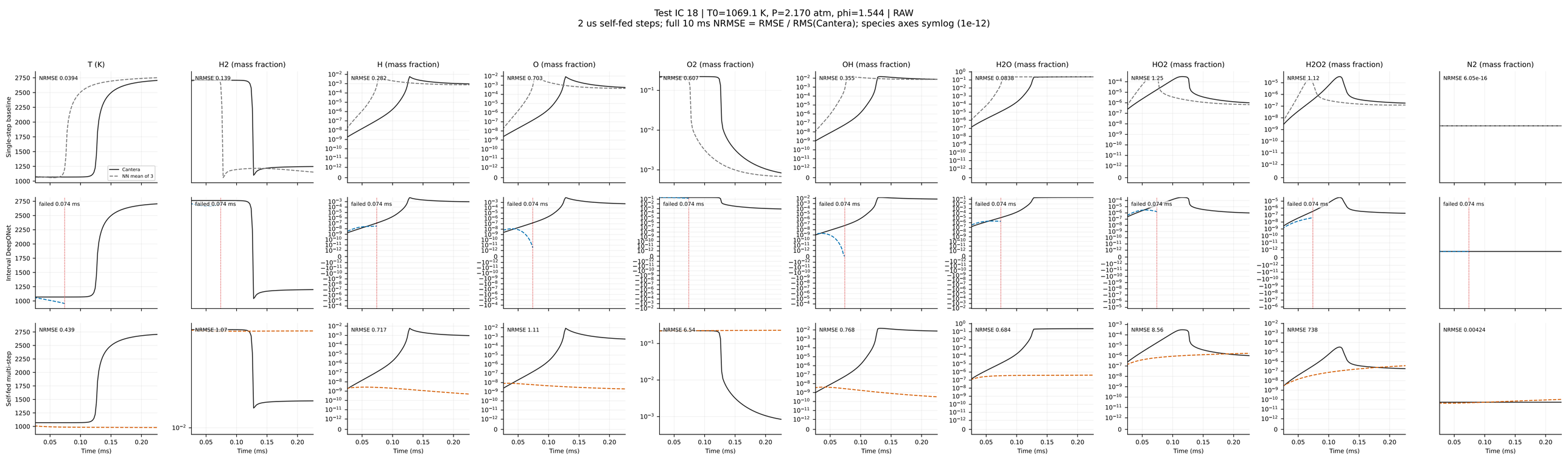
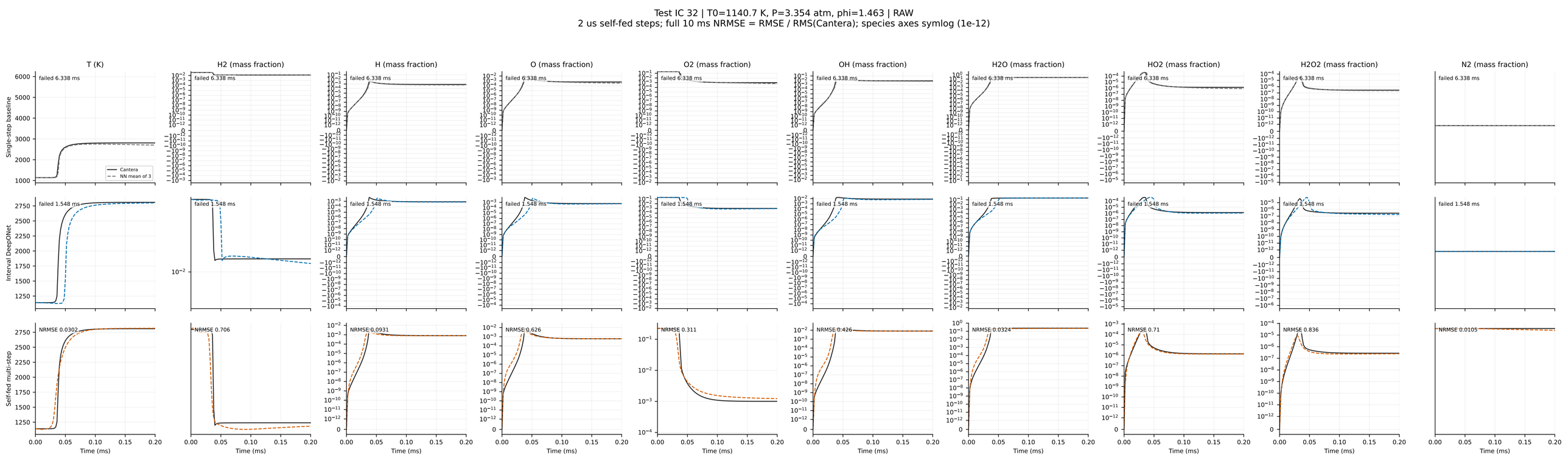
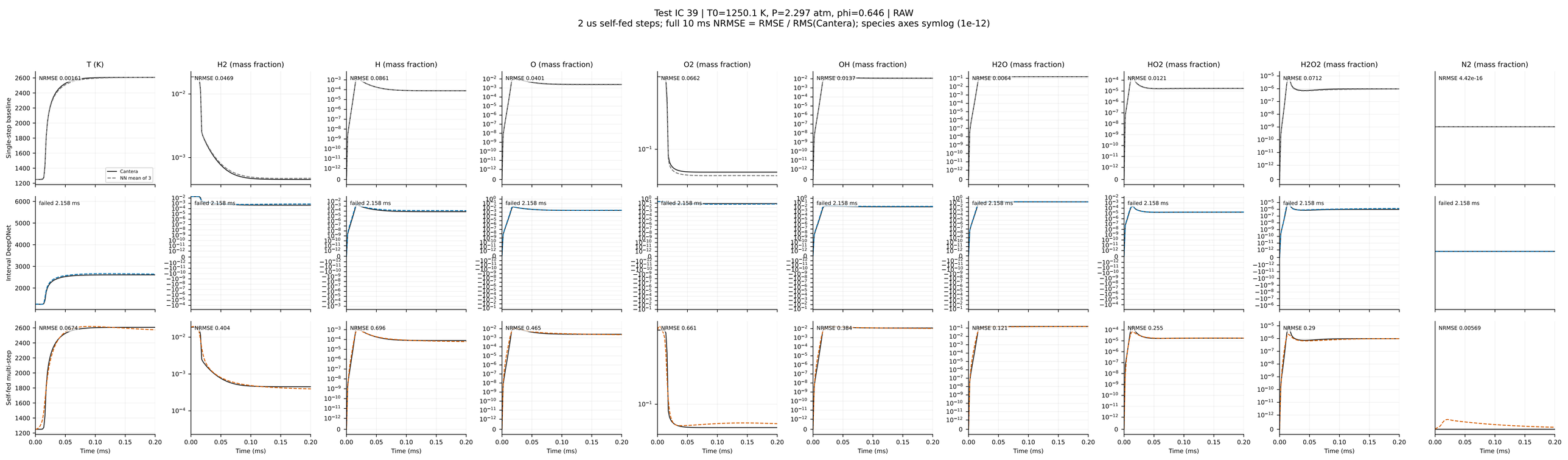

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0736
NRMSE 0.0423
NRMSE 0.292
NRMSE 0.573
NRMSE 0.0246
NRMSE 0.452
NRMSE 0.0699
NRMSE 0.123
NRMSE 0.959
NRMSE 0
Cantera
NN mean of 3
failed 1.122 ms
NRMSE 0.0179
NRMSE 0.103
NRMSE 0.0746
NRMSE 0.276
NRMSE 0.503
NRMSE 0.136
NRMSE 0.0389
NRMSE 0.862
NRMSE 0.583
NRMSE 0.00173
Time (ms)

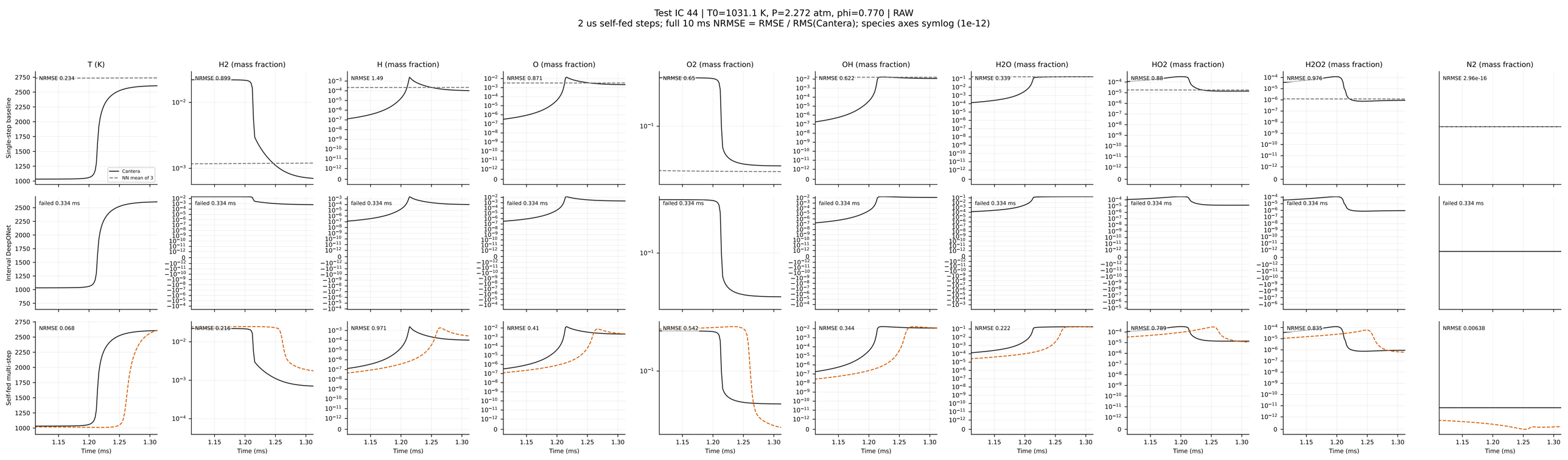

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.066 ms
failed 0.02 ms
failed 0 ms
Cantera
NN mean of 3
Time (ms)

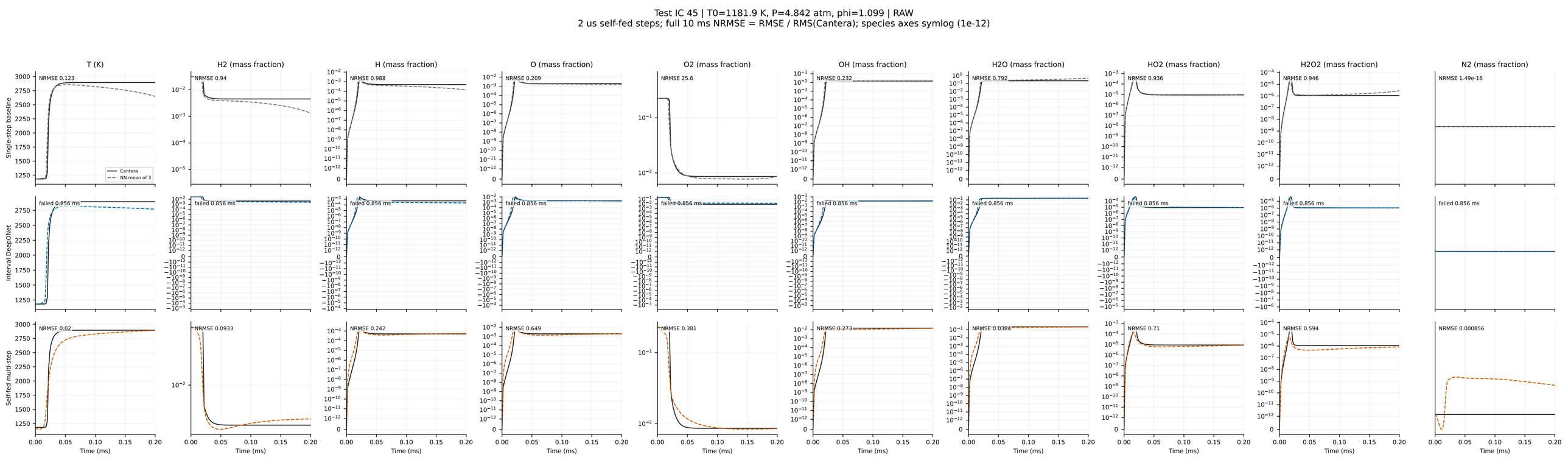

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 1.076 ms
failed 0.034 ms
failed 0 ms
Cantera
NN mean of 3
+9.306e2
Time (ms)

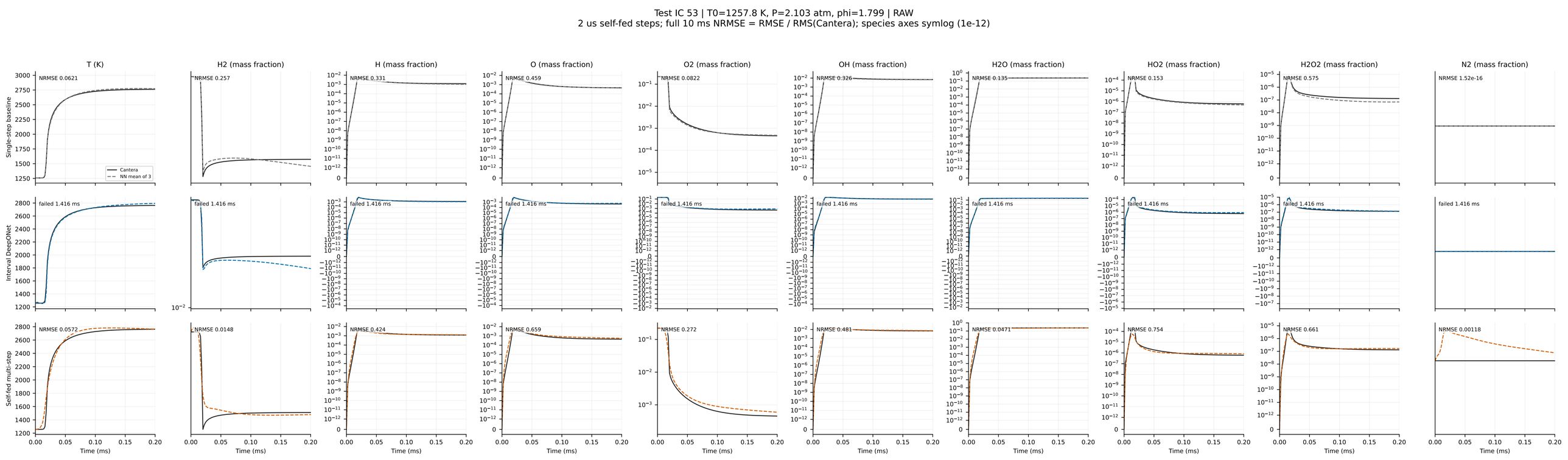

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0394
NRMSE 0.139
NRMSE 0.282
NRMSE 0.703
NRMSE 0.607
NRMSE 0.355
NRMSE 0.0838
NRMSE 1.25
NRMSE 1.12
NRMSE 6.05e-16
Cantera
NN mean of 3
failed 0.074 ms
NRMSE 0.439
NRMSE 1.07
NRMSE 0.717
NRMSE 1.11
NRMSE 6.54
NRMSE 0.768
NRMSE 0.684
NRMSE 8.56
NRMSE 738
NRMSE 0.00424
Time (ms)

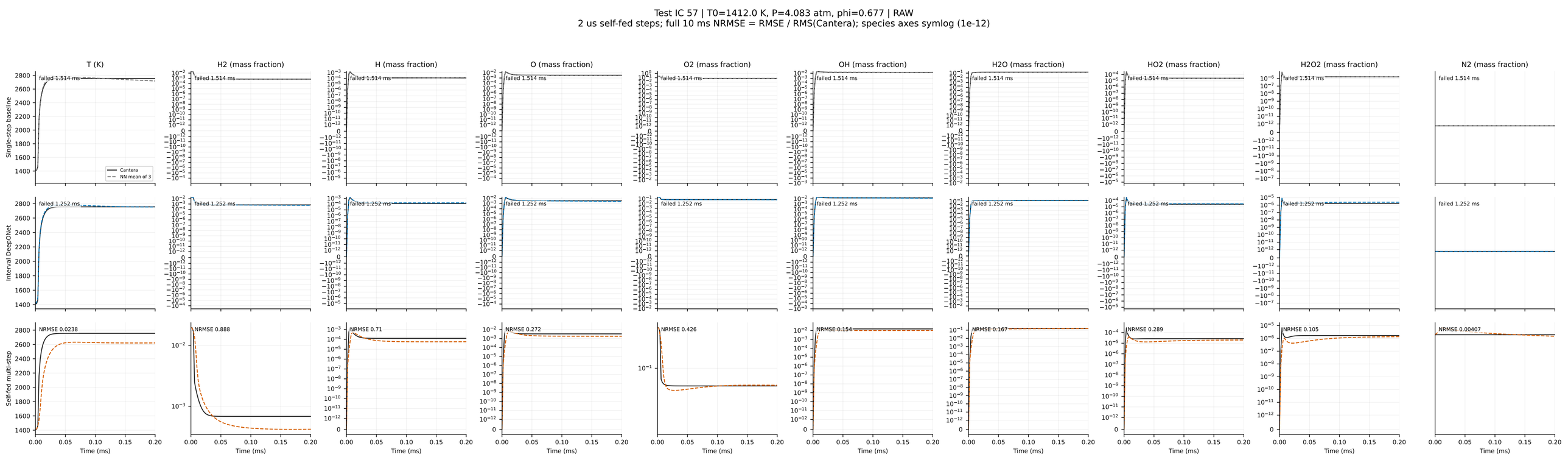

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 6.338 ms
failed 1.548 ms
Cantera
NN mean of 3
NRMSE 0.0302
NRMSE 0.706
NRMSE 0.0931
NRMSE 0.626
NRMSE 0.311
NRMSE 0.426
NRMSE 0.0324
NRMSE 0.71
NRMSE 0.836
NRMSE 0.0105
Time (ms)

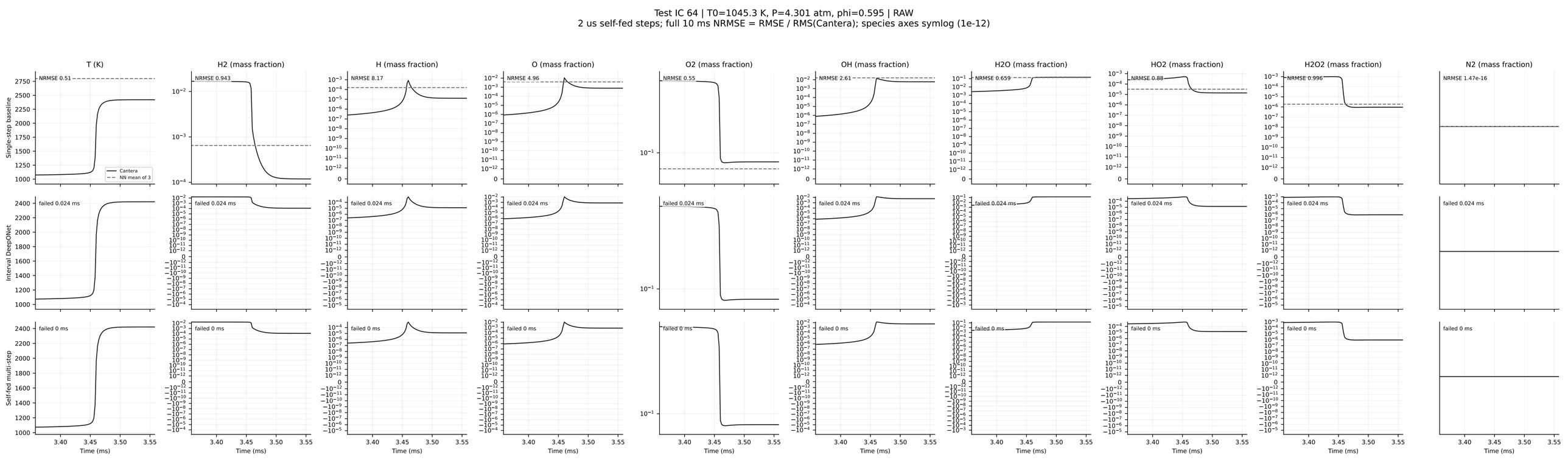

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.00161
NRMSE 0.0469
NRMSE 0.0861
NRMSE 0.0401
NRMSE 0.0662
NRMSE 0.0137
NRMSE 0.0064
NRMSE 0.0121
NRMSE 0.0712
NRMSE 4.42e-16
failed 2.158 ms
NRMSE 0.0674
NRMSE 0.404
NRMSE 0.696
NRMSE 0.465
NRMSE 0.661
NRMSE 0.384
NRMSE 0.121
NRMSE 0.255
NRMSE 0.29
NRMSE 0.00569
Cantera
NN mean of 3
Time (ms)

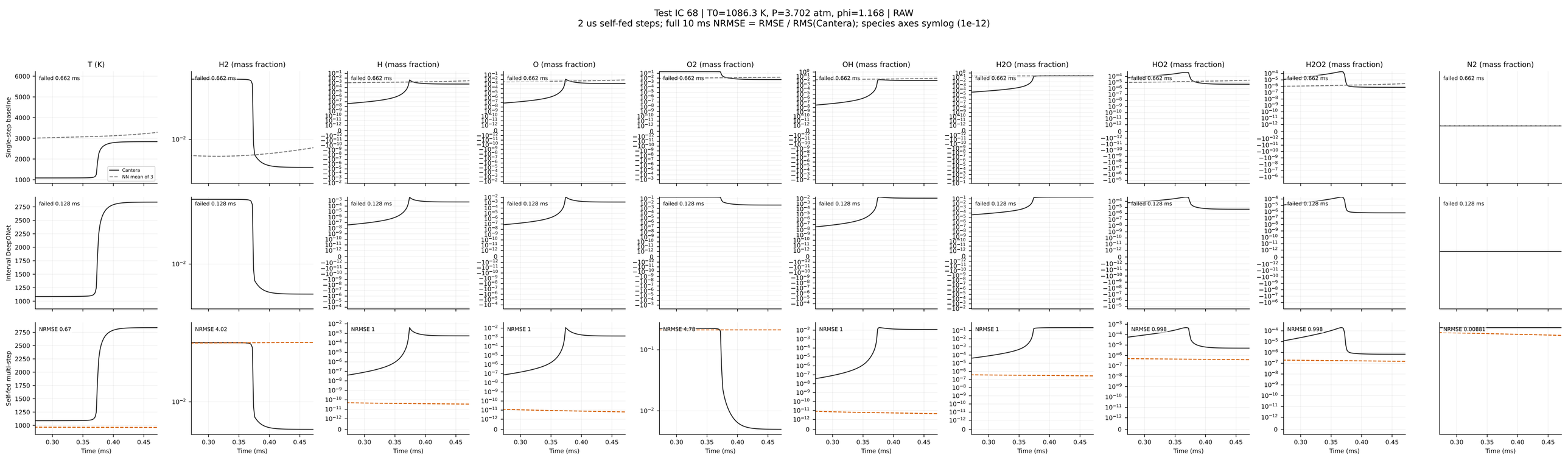

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.234
NRMSE 0.899
NRMSE 1.49
NRMSE 0.871
NRMSE 0.65
NRMSE 0.622
NRMSE 0.339
NRMSE 0.88
NRMSE 0.976
NRMSE 2.96e-16
Cantera
NN mean of 3
failed 0.334 ms
NRMSE 0.068
NRMSE 0.216
NRMSE 0.971
NRMSE 0.41
NRMSE 0.542
NRMSE 0.344
NRMSE 0.222
NRMSE 0.789
NRMSE 0.835
NRMSE 0.00638
Time (ms)

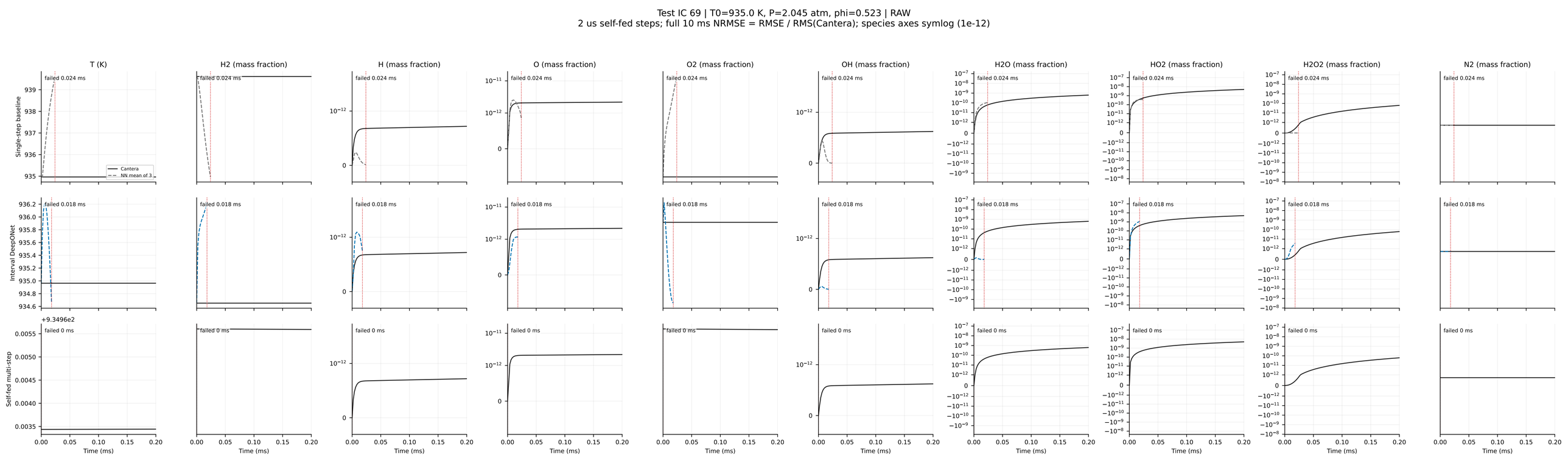

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.123
NRMSE 0.94
NRMSE 0.988
NRMSE 0.209
NRMSE 25.6
NRMSE 0.232
NRMSE 0.792
NRMSE 0.936
NRMSE 0.946
NRMSE 1.49e-16
failed 0.856 ms
NRMSE 0.02
NRMSE 0.0933
NRMSE 0.242
NRMSE 0.649
NRMSE 0.381
NRMSE 0.273
NRMSE 0.0384
NRMSE 0.71
NRMSE 0.594
NRMSE 0.000856
Cantera
NN mean of 3
Time (ms)

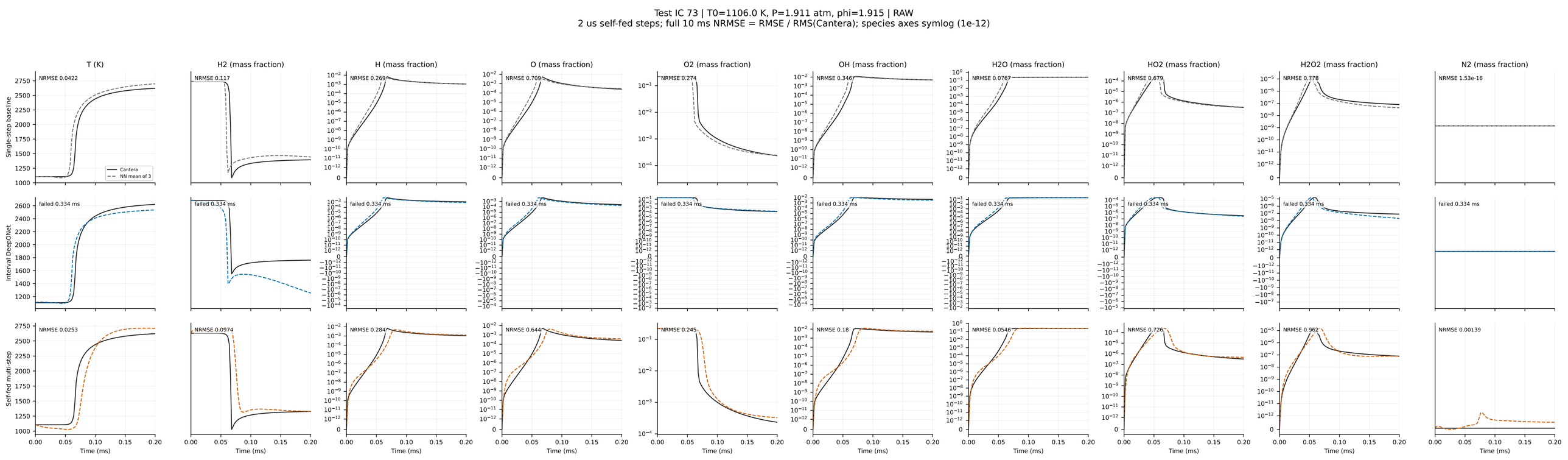

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0621
NRMSE 0.257
NRMSE 0.331
NRMSE 0.459
NRMSE 0.0822
NRMSE 0.326
NRMSE 0.135
NRMSE 0.153
NRMSE 0.575
NRMSE 1.52e-16
failed 1.416 ms
NRMSE 0.0572
NRMSE 0.0148
NRMSE 0.424
NRMSE 0.659
NRMSE 0.272
NRMSE 0.481
NRMSE 0.0471
NRMSE 0.754
NRMSE 0.661
NRMSE 0.00118
Time (ms)

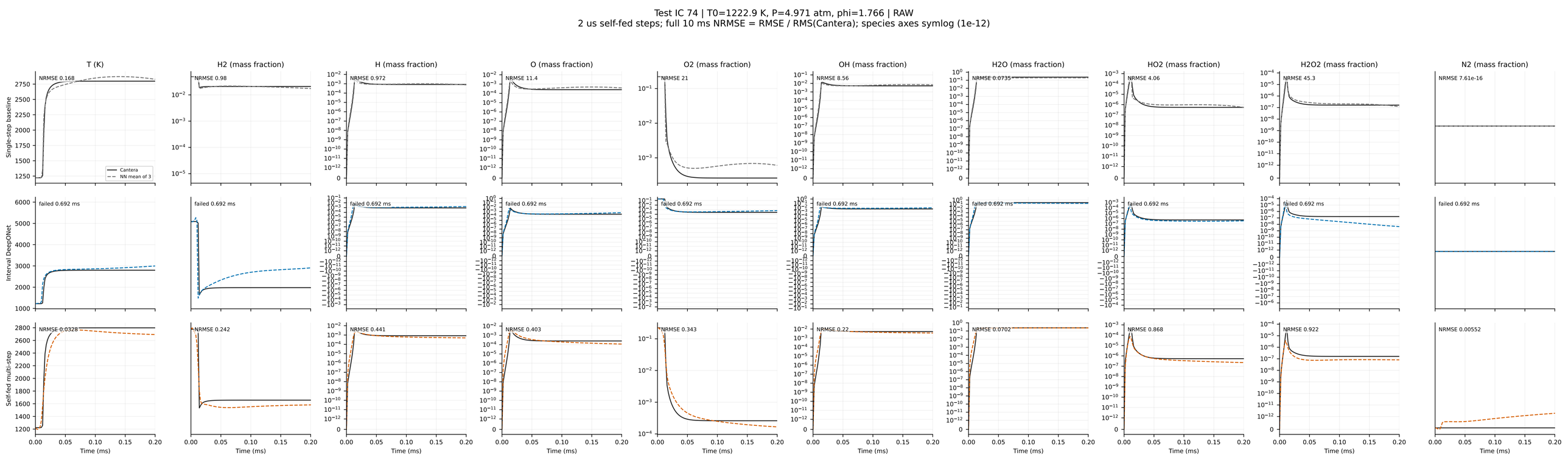

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 1.514 ms
failed 1.252 ms
NRMSE 0.0238
NRMSE 0.888
NRMSE 0.71
NRMSE 0.272
NRMSE 0.426
NRMSE 0.154
NRMSE 0.167
NRMSE 0.289
NRMSE 0.105
NRMSE 0.00407
Time (ms)

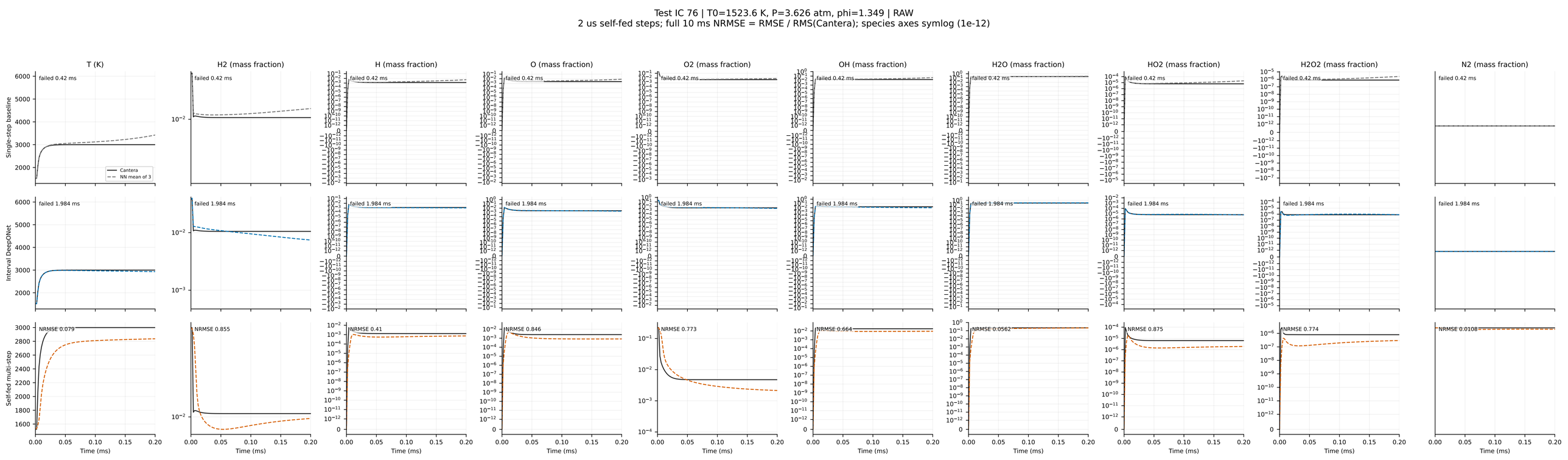

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.51
NRMSE 0.943
NRMSE 8.17
NRMSE 4.96
NRMSE 0.55
NRMSE 2.61
NRMSE 0.659
NRMSE 0.88
NRMSE 0.996
NRMSE 1.47e-16
Cantera
NN mean of 3
failed 0.024 ms
failed 0 ms
Time (ms)

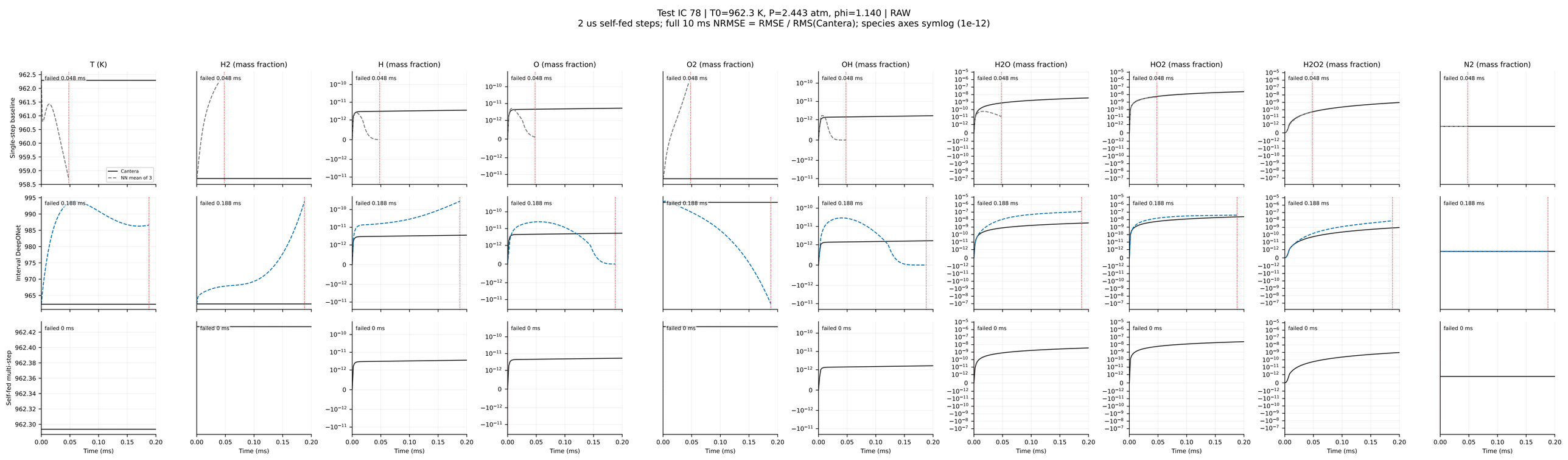

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.662 ms
failed 0.128 ms
NRMSE 0.67
NRMSE 4.02
NRMSE 1
NRMSE 4.78
NRMSE 0.998
NRMSE 0.00881
Cantera
NN mean of 3
Time (ms)

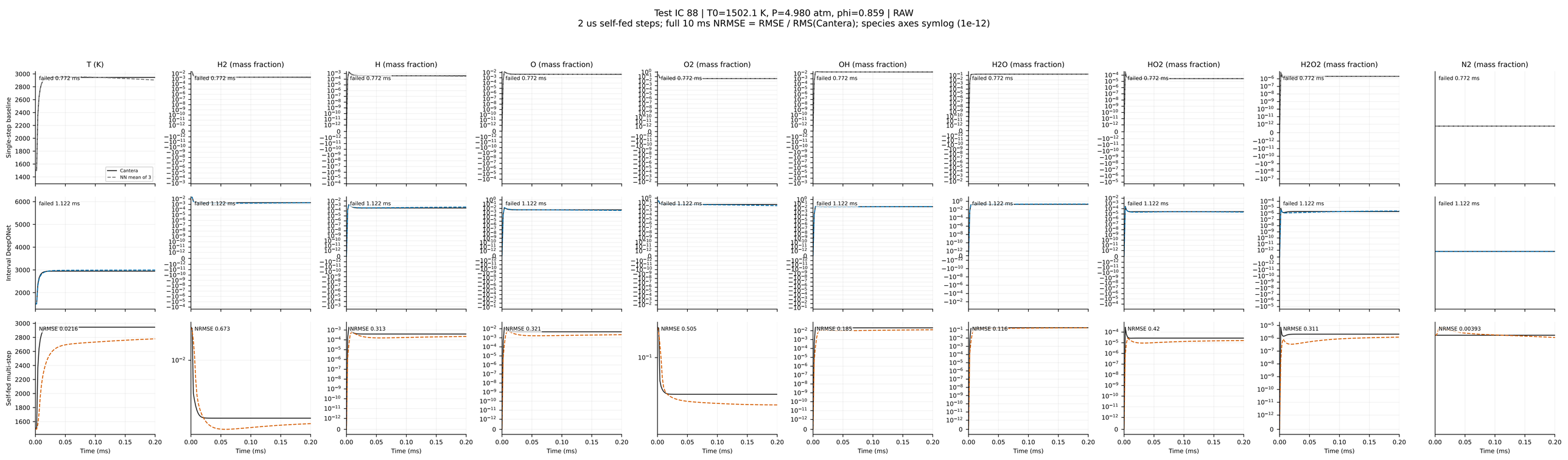

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.024 ms
failed 0.018 ms
failed 0 ms
Cantera
NN mean of 3
+9.3496e2
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0422
NRMSE 0.117
NRMSE 0.269
NRMSE 0.709
NRMSE 0.274
NRMSE 0.346
NRMSE 0.0767
NRMSE 0.679
NRMSE 0.778
NRMSE 1.53e-16
failed 0.334 ms
NRMSE 0.0253
NRMSE 0.0974
NRMSE 0.284
NRMSE 0.644
NRMSE 0.245
NRMSE 0.18
NRMSE 0.0546
NRMSE 0.726
NRMSE 0.962
NRMSE 0.00139
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.168
NRMSE 0.98
NRMSE 0.972
NRMSE 11.4
NRMSE 21
NRMSE 8.56
NRMSE 0.0735
NRMSE 4.06
NRMSE 45.3
NRMSE 7.61e-16
failed 0.692 ms
NRMSE 0.0328
NRMSE 0.242
NRMSE 0.441
NRMSE 0.403
NRMSE 0.343
NRMSE 0.22
NRMSE 0.0702
NRMSE 0.868
NRMSE 0.922
NRMSE 0.00552
Cantera
NN mean of 3
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.42 ms
failed 1.984 ms
Cantera
NN mean of 3
NRMSE 0.079
NRMSE 0.855
NRMSE 0.41
NRMSE 0.846
NRMSE 0.773
NRMSE 0.664
NRMSE 0.0562
NRMSE 0.875
NRMSE 0.774
NRMSE 0.0108
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.048 ms
failed 0.188 ms
failed 0 ms
Cantera
NN mean of 3
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 0.772 ms
failed 1.122 ms
NRMSE 0.0216
NRMSE 0.673
NRMSE 0.313
NRMSE 0.321
NRMSE 0.505
NRMSE 0.185
NRMSE 0.116
NRMSE 0.42
NRMSE 0.311
NRMSE 0.00393
Time (ms)